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PHYSICS GRADE 9-12 FULL DETAILED REVISION

A complete, comprehensive analytical guide strictly aligned with the EUEE syllabus. This guide covers all fundamental theories, crucial formulas, and includes exactly **50 embedded multiple-choice questions from the 2018 EUEE Physics Exam** formatted for quick review.

Chapter 1: Application of Physics

Physics is the foundational science that explains how the universe behaves. Its applications permeate every aspect of modern society, from the structures we build to the medical tools that save lives.

1.1 - 1.5: Society, Communities, and Knowledge Making

- **Importance to Society (1.1):** Physics drives technological advancement, economic development, and provides solutions to global challenges (like renewable energy).

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Prepared by @NextGen_12 | Ethiopian University Entrance Exam (EUEE) Physics Note

interested physics professionals primarily to foster education, contribute to academic research, and build networks of shared knowledge. They do not directly enforce government directives.

- **Making of Physics Knowledge (1.3):** Involves the scientific method: Observation → Hypothesis → Experimentation → Theory → Law.
- **Mission & Career Awareness (1.4):** Physics graduates work in education, research (CERN, NASA), finance, data science, and telecommunications.
- **Current Status of Physics (1.5):** Modern physics is heavily focused on Quantum Mechanics, Relativity, String Theory, and finding a Grand Unified Theory.

1.6 - 1.10: Intersections with Other Disciplines

- **Physics and Other Sciences (1.6):** Biophysics (physics in biology), Physical Chemistry, and Astrophysics are prime examples of interdisciplinary studies.
- **Physics and Engineering (1.7):** Engineering is applied physics. Civil engineering uses statics and dynamics; electrical engineering uses electromagnetism.
- **Medical Physics (1.8):** Utilizes radiation and waves. Examples include **Ultrasound imaging** (using high-frequency sound waves where anechoic tissues appear black because they don't reflect sound back), X-rays for bone structure, MRI (magnetic resonance), and radioactive isotopes (like Iodine-131) for treatment.
- **Defense Technology (1.9):** Physics is crucial in radar systems, satellite navigation (GPS relies on relativity), and ballistics.
- **Communication (1.10):** Fiber optics (total internal reflection of light) and radio waves are foundational for internet and telecommunications.

2018 EUEE QUESTIONS - CHAPTER 1

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appear on the screen as:

- A. black because the tissues do not reflect any sound wave.
- B. light gray because the tissues do not reflect any sound wave.
- C. black because the tissues reflect intense sound waves.
- D. light gray because the tissues reflect intense sound waves.

Answer: A

Anechoic means "free from echoes." Without reflection, no signal returns, rendering a black image on the ultrasound display.

14. Ethiopian Physical Society is established by interested Ethiopian physics professionals to:

- A. prepare physics curriculum under the ministry of education directives.
- B. contribute its own share to promote physics education and research.
- C. take part in election and participate in government formation.
- D. monitor and supervise the curriculum implementation in high schools.

Answer: B

Professional societies are non-governmental; their primary aim is advancing research and promoting education.

advancement of physics correctly?

A. The invention of space shuttles enabled experiments in space.

B. The development of light detectors enabled spectral analysis.

C. The invention of x-rays enabled proper medical imaging.

D. The development of x-rays enabled study of atomic structure.

Answer: C

Medical imaging is an application of physics TO medicine, rather than technology contributing to the foundational study of physics itself.

Chapter 2: Vectors

Physical quantities are either **Scalars** (magnitude only, e.g., mass, time, temperature, distance) or **Vectors** (magnitude and direction, e.g., velocity, acceleration, force, displacement).

2.1 - 2.5: Vector Representation, Addition & Resolution

- **Vector Representations (2.2):** Vectors are represented graphically by arrows. The **length** of the arrow corresponds strictly to the vector's magnitude according to a chosen scale, while the **head (arrowhead)** represents the direction.
- **Graphical Method of Vector Addition (2.4):** Includes the Triangle Law (head-to-tail method) and Parallelogram Law (tail-to-tail method). **Co-planar vectors** are vectors that exist within the exact same 2D plane.
- **Vector Resolution (2.5):** Splitting a vector into perpendicular components. For a vector A at angle θ with the x-axis: $A_x = A \cos(\theta)$ and $A_y = A \sin(\theta)$.

2.6: Product of Vectors

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- **Dot Product (Scalar Product):** $A \cdot B = AB \cos(\theta)$. The result is a scalar. It is used to find Work ($W = F \cdot d$).
- **Cross Product (Vector Product):** $A \times B = AB \sin(\theta) n$. The result is a new vector perpendicular to the plane of A and B, determined by the Right-Hand Rule. It is used to find Torque ($\tau = r \times F$).

2018 EUEE QUESTIONS - CHAPTER 2

15. If an arrow is drawn to a scale to describe a given vector, then its:

- A. length corresponds to the vector magnitude.
- B. tail stands for the terminal point of the vector.
- C. length corresponds to the vector direction.
- D. head represents the initial point of the vector.

Answer: A

In vector graphics, the scaled length denotes the magnitude.

37. If a student walks 100 m due 37° North of East and then 120 m towards South, what is the displacement moved by the student?

- A. 220 m, 53° North of East
- B. 100 m, 53° North of East
- C. 220 m, 37° South of East
- D. 100 m, 37° South of East

Answer: D

Resolve V1: $100\cos(37^\circ)\mathbf{i} + 100\sin(37^\circ)\mathbf{j} = 80\mathbf{i} + 60\mathbf{j}$. V2: $-120\mathbf{j}$. Resultant = $80\mathbf{i} - 60\mathbf{j}$. Magnitude = $\sqrt{(80^2 + (-60)^2)} = 100\text{m}$. Direction is South of East

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A. lying in the same plane.

B. perpendicular and lying in two parallel planes.

C. parallel and lying in two parallel planes.

D. lying in different planes.

Answer: A

"Co-" means together/same, and "planar" refers to a plane.

Chapter 3: Kinematics

Kinematics is the description of motion without regard to the forces causing it. It focuses on position, displacement, speed, velocity, and acceleration across straight lines, curves, and circles.

3.1 - 3.7: 1D Motion, Graphs & Constant Acceleration

- **Distance vs Displacement (3.1):** Distance is the total path covered (scalar). Displacement is the straight line from start to finish (vector). On any curved path, distance is strictly greater than the magnitude of displacement.
- **Speed vs Velocity (3.2 & 3.3):** Average speed = Total Distance / Total Time. Average Velocity = Net Displacement / Total Time. Instantaneous velocity is the velocity at a specific point in time (the derivative of position with respect to time).
- **Graphical Representations (3.6):** The slope of a position-time graph gives velocity. The slope of a velocity-time graph gives acceleration. The area under a velocity-time graph gives displacement.
- **Vertical Motion (3.7):** Objects in free fall experience constant acceleration due to gravity ($g \approx 9.8$ or 10 m/s^2 downwards).

$$\mathbf{v} = \mathbf{v}_0 + \mathbf{a}t$$

$$\mathbf{x} = \mathbf{x}_0 + \mathbf{v}_0 t + 0.5 \mathbf{a} t^2$$

$$\mathbf{v}^2 = \mathbf{v}_0^2 + 2\mathbf{a}(\mathbf{x} - \mathbf{x}_0)$$

$$\mathbf{x} = \mathbf{x}_0 + [(\mathbf{v} + \mathbf{v}_0) / 2] t$$

3.8 - 3.13: 2D & Rotational Kinematics

- **Projectile Motion (3.9):** Motion in two dimensions under the influence of gravity alone. The horizontal velocity is constant ($a_x = 0$), while vertical motion undergoes constant acceleration ($-g$). The path is a parabola.
- **Uniform Circular Motion (3.8):** On a horizontal plane, the tangential speed is constant. However, because direction is constantly changing, the tangential velocity vector is not constant, meaning there is a non-zero centripetal acceleration ($a_c = v^2/r$) acting towards the center.
- **Angular Motion (3.10 - 3.13):** Angular position (θ), Angular velocity ($\omega = \Delta\theta/\Delta t$), and Angular acceleration ($\alpha = \Delta\omega/\Delta t$). The relation between linear and angular components: $\mathbf{v} = \mathbf{r}\omega$ and $\mathbf{a}_t = \mathbf{r}\alpha$. By the Right-Hand Rule, ω and α point perpendicularly out of or into the plane of rotation.

2018 EUEE QUESTIONS - CHAPTER 3

13. Consider a car performing a motion with constant acceleration (a) starting from position x_0 at time $t = 0$ with initial velocity v_0 . Which equation correctly governs the motion?

A. $x = -x_0 + [2(v+v_0)/t]$

B. $x = x_0 + [(v+v_0)/2]t$

C. $x = x_0 + [v^2+v_0^2]/(2a)$

D. $x = x_0 + [(v-v_0)/2]t$

Answer: B

Average velocity = $(v+v_0)/2$. Displacement = avg_velocity * time. So, $x - x_0 = [(v+v_0)/2]t$.

velocity of car A with respect to the ground is 30 km/hr towards East, what would be the velocity of car B with respect to the ground?

A. 50 km/hr towards West

B. 50 km/hr towards East

C. 110 km/hr towards East

D. 110 km/hr towards West

Answer: A

$V_{B/A} = V_B - V_A$. Let East be positive (+). $V_{B/A} = -80$. $V_A = +30$. $-80 = V_B - 30 \Rightarrow V_B = -50$ (50 km/hr West).

22. Which one of the following statements correctly describes the rotation of an object around a fixed axis?

A. Angular accel is perpendicular to the plane of rotation and in same direction to angular velocity if rotation rate is increasing.

B. Particles closer to axis have different angular displacement.

C. Particles closer to axis have larger angular velocity.

D. Particles closer to axis have similar translational displacement.

Answer: A

By the Right-Hand Rule, angular vectors are perpendicular to the plane. If speeding up, α and ω point the same way.

constant velocity. It then decelerates uniformly with 1 m/s^2 until it stops after 200 m. What is the average velocity for the total trip?

A. 20.0 m/s

B. 6.5 m/s

C. 13.3 m/s

D. 10.0 m/s

Answer: C

Total Distance = 1200m. Max speed from decel phase: $v^2 = 2(1)(200) \Rightarrow v = 20 \text{ m/s}$. Times: $t_1 = 400/10 = 40\text{s}$. $t_2 = 600/20 = 30\text{s}$. $t_3 = 200/10 = 20\text{s}$. Total Time = 90s. Avg $V = 1200/90 = 13.3 \text{ m/s}$.

43. Which one of the following statements correctly describes a uniform circular motion of a particle in a horizontal plane? Its:

A. centripetal acceleration is a non-zero constant.

B. tangential velocity is a non-zero constant.

C. angular velocity is a non-zero constant.

D. tangential acceleration is a non-zero constant.

Answer: C

Tangential velocity direction changes (so it's not constant). Centripetal acceleration direction changes (always points to center). Angular velocity vector remains constant along the axis of rotation.

A. its velocity changes equally with equal interval of time.

B. it moves with constant speed.

C. it moves with constant velocity.

D. its position changes equally with equal interval of time.

Answer: A

Acceleration is the rate of change of velocity. If constant, the rate of change is equal over equal times.

50. For an athlete who runs on a track for a specified time interval, her distance and displacement can be different in that distance:

A. can be negative whereas displacement cannot be.

B. is less than the magnitude of displacement.

C. can be zero whereas the displacement cannot be.

D. is greater than the magnitude of displacement.

Answer: D

Distance is the actual path length, which is always equal to or greater than the straight-line displacement magnitude.

Chapter 4: Force, Work and Energy

Newton's Laws connect kinematics to mass, detailing the causes of motion. The principles of Energy and Momentum allow us to solve complex physical states.

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- **Newton's Laws of Motion (4.2):** (1) Inertia, (2) $F = ma$, (3) Action/Reaction pairs.
- **Forces of Friction (4.3):** Static friction (prevents slipping, $F_s \leq \mu_s N$) is generally greater than kinetic friction (sliding, $F_k = \mu_k N$).
- **Impulse and Linear Momentum (4.4):** Impulse ($J = F\Delta t$) equals change in momentum (Δp). In any isolated system, total momentum is conserved.
- **Collisions:** *Elastic Collisions:* Both total kinetic energy and total momentum are conserved (e.g., ideal billiard balls). *Inelastic Collisions:* Momentum is conserved, but kinetic energy is lost to heat/sound/deformation.

4.5 - 4.9: Rotational Dynamics & Equilibrium

- **Center of Mass (4.5):** The point where the entire mass of an object is assumed to be concentrated for translational motion.
- **Torque and Rotational Inertia (4.6):** Torque ($\tau = rF \sin\theta$) causes angular acceleration. Moment of inertia (I) is the rotational equivalent of mass, dependent on mass distribution relative to the axis.
- **Rotational Kinetic Energy & Momentum (4.7 & 4.8):** Rotational KE = $\frac{1}{2} I\omega^2$. Angular Momentum ($L = I\omega$) is conserved if net external torque is zero.
- **Equilibrium of Rigid Bodies (4.9):** Requires net force = 0 (translational equilibrium) AND net torque = 0 (rotational equilibrium).

4.10 - 4.14: Work, Energy, and Power

- **Work (4.10):** Work is defined as the dot product of force and displacement ($W = F \cdot d$). For variable forces, work is the area under the Force-Distance graph.
- **Work-Energy Theorem (4.12):** Net work done on an object equals its change in Kinetic Energy ($W_{\text{net}} = \Delta KE$).
- **Conservative Forces (4.13):** A conservative force (like gravity, electric force, or spring force) is unique because the work done by it is strictly *path independent*—it only matters where you start and end. Work done by a conservative force along a closed path is exactly zero.
- **Power (4.14):** The rate at which work is done ($P = W/t$ or $P = F \cdot v$).

2018 EUEE QUESTIONS - CHAPTER 4

5. A 25 kg box is dragged on a horizontal plane from $r_1 = (1.3\mathbf{i} + 1.4\mathbf{j})$ m to $r_2 = (8.3\mathbf{i} + 2.9\mathbf{j})$ m by a force $F = (4\mathbf{i} - 6\mathbf{j})$ N. How much work is done?

A. 19.0 J

B. 37.0 J

C. 15.8 J

D. 75.6 J

Answer: A

$$\Delta r = (7.0\mathbf{i} + 1.5\mathbf{j}). W = F \cdot \Delta r = (4)(7) + (-6)(1.5) = 28 - 9.0 = 19.0 \text{ J.}$$

8. A constant force of 250.0 N is applied horizontally on a box on a frictionless surface. If it accelerates at 4.0 m/s^2 , what is its mass?

A. 125.0 kg

B. 41.7 kg

C. 10.4 kg

D. 62.5 kg

Answer: D

$$F = ma \Rightarrow m = F/a = 250.0 / 4.0 = 62.5 \text{ kg.}$$

A is raised to $h/2$, which is correct about their gravitational potential energies?

A. A is equal to that of B.

B. A is twice that of B.

C. B is twice that of A.

D. A is 4 times that of B.

Answer: A

$PE_A = (2m)(g)(h/2) = mgh$. $PE_B = (m)(g)(h) = mgh$. They are equal.

35. Conservative force differs from the nonconservative one in that, the work done by:

A. conservative force along a closed path is non-zero...

B. conservative force is path independent unlike that by non-conservative one.

C. non-conservative force is path independent...

D. non-conservative force along a closed path is zero...

Answer: B

Conservative forces (gravity, electrostatic, spring) do work independent of the path taken.

and place it on a bookshelf 2.0 m above the floor, the work done by you and by gravity, respectively, are:

A. -60 J and 60 J

B. 33 J and -33 J

C. -33 J and 33 J

D. 60 J and -60 J

Answer: B

$\Delta h = 2.0 - 0.9 = 1.1\text{m}$. $W_{\text{you}} = mgh = 3 \times 10 \times 1.1 = 33\text{ J}$. Gravity opposes this, so $W_{\text{gravity}} = -33\text{ J}$.

51. A collision between two particles is known as:

A. elastic if the total kinetic energy is not conserved...

B. elastic if both the total kinetic energy and momentum are conserved.

C. inelastic if the total kinetic energy is conserved...

D. inelastic if both...

Answer: B

Elastic collisions uniquely conserve Kinetic Energy alongside momentum.

Chapter 5: Simple Machines

Machines do not reduce the amount of work done; they reduce the *effort force* required by increasing the distance over which that force is applied.

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- **Classification of Simple Machines (5.3):** The classical simple machines are: **Lever, Pulley, Wheel and axle, Inclined plane, Wedge, and Screw**. Complex items like car engines, gas cylinders, and seat belts are not simple machines.
- **Levers:** Classified by what is in the middle (FLE = 123):
 - First Order: **Fulcrum** in middle (e.g., seesaw, scissors).
 - Second Order: **Load** in middle (e.g., wheelbarrow, nutcracker).
 - Third Order: **Effort** in middle (e.g., tweezers, human forearm). The effort distance is less than load distance.
- **Designing Simple Machines (5.5):** Combining components to increase mechanical advantage while accounting for frictional losses.

Mechanical Advantage (MA) = Load / Effort

Velocity Ratio (VR) = Distance moved by Effort / Distance moved by Load

Efficiency (η) = (MA / VR) $\times$ 100% (Efficiency is always $< 100\%$ due to friction)

For a pulley system, ideal MA = number of load-bearing rope strands.

2018 EUEE QUESTIONS - CHAPTER 5

16. From the given lists of machines, which one contains only simple machines?

A. Gas-cylinder, Wedge, Screw, Lever

B. Compound pulley, Wheel and axle, Seat belt, Inclined plane

C. Washing machine, Car engine, Wedge, Lever

D. Inclined plane, Screw, Wheel and axle, Wedge

Answer: D

All listed in D are fundamental classical simple machines

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A. The load is between the effort and fulcrum.

B. The fulcrum is between the load and the effort.

C. The effort distance is greater than load distance.

D. The effort is between the load and the fulcrum.

Answer: D

In class 3 levers (like tweezers or the human forearm), the effort is applied between the fulcrum and the load.

38. A compound pulley system consisting of two moveable and two fixed pulleys is used to lift a 400 N object... If frictional force is negligible, what is the mechanical advantage?

A. 4

B. 2

C. 5

D. 1

Answer: A

In an ideal block and tackle pulley system with 4 supporting rope segments holding the moveable block, the MA is 4.

Chapter 6 & 7: Temperature, Heat & Elasticity

Thermodynamics studies heat transfer, temperature scaling, and expansion, while solid mechanics (elasticity) studies how rigid bodies deform under stress.

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- **Heat vs Temperature (6.1 & 6.3):** Temperature is a measure of average kinetic energy. **Heat** is exclusively defined as the form of energy that naturally transfers when two objects of different temperatures are brought into thermal contact.
- **Thermometric Scales & Conversions (6.4 & 6.6):** Celsius, Fahrenheit, and Kelvin. Formula: $C / 5 = (F - 32) / 9 = (K - 273.15) / 5$.
- **Types of Thermometers (6.5):** Liquid-in-glass (mercury/alcohol), thermocouples (utilizing voltage differences across metal junctions), and bimetallic strips.
- **Determining Specific Heat Capacity:** One can use the mixture method (calorimetry) or the **electrical method**, where heat is supplied to a solid for a known time by an electrical heater of known power ($E = Pt = mc\Delta T$).
- **Thermal Expansion (6.7):** Materials expand when heated. Coefficients of expansion relate strictly: Surface/Area (β) = 2α and Volume (γ) = 3α , where α is the linear expansion coefficient.

7.1: Elasticity and Static Equilibrium

- When external forces deform an object, we define:
Stress = Force / Cross-Sectional Area (Causes the deformation, measured in N/m^2 or Pascals).
Strain = $\Delta \text{Length} / \text{Original Length}$ (The fractional change describing the deformation itself, unitless).
- **Young's Modulus (Y)** = Stress / Strain = $(F L) / (A \Delta L)$. It is a measure of the stiffness of a solid material.

2018 EUEE QUESTIONS - CH 6 & 7

120 kg mass is attached... Young's modulus is $6 \times 10^{10} \text{ N/m}^2$. Change in length?

A. 0.05 m

B. 0.04 m

C. 0.06 m

D. 0.02 m

Answer: B

$$\Delta L = (mgL)/(AY) = (1200 \times 6) / (3 \times 10^{-6} \times 6 \times 10^{10}) = 7200 / 180000 = 0.04 \text{ m.}$$

3. If coefficients of thermal expansion are α (linear), β (surface), and γ (volume), they relate as:

A. $\gamma = 2\alpha$

B. $2\beta = 3\gamma$

C. $3\beta = 2\gamma$

D. $\beta = 3\alpha$

Answer: C

$$\beta = 2\alpha \text{ and } \gamma = 3\alpha. \text{ Thus } \alpha = \beta/2 = \gamma/3 \Rightarrow 3\beta = 2\gamma.$$

length or volume is:

A. stress.

B. strain.

C. elasticity.

D. density.

Answer: B

Strain = $\Delta L/L$ (a fractional change).

41. Which one of the following statements describes heat?

A. It is the form of energy that transfers when two objects of different temperatures are brought in thermal contact.

B. It is the form of energy which is stored...

C. It is a property related to average kinetic energy...

D. It is a form of energy doing work...

Answer: A

Heat is exclusively defined as energy in transit driven by a ΔT .

A. mixture method with heat energy supplied...

B. electrical method with the heat energy supplied for a known time by an electrical heater of known power.

C. electrical method with hot water...

D. mixture method with unknown time...

Answer: B

Electrical method equates electrical work ($P \cdot t$) to heat gained ($mc\Delta T$).

Chapter 8: Fluid Mechanics

This chapter studies fluids at rest (statics) and in motion (dynamics).

8.1 - 8.4: Fluid Statics & Buoyancy

- **Pressure (8.1 & 8.2):** The pressure in a fluid at rest depends ONLY on the depth (h) and the density of the fluid (ρ), given by $P = \rho gh$. The shape of the container does not matter. *Pascal's Principle* states that pressure applied to an enclosed fluid is transmitted undiminished (basis of hydraulic lifts).
- **Atmospheric Pressure (8.3):** Measured using Barometers (Torricelli's invention using mercury). Standard atm = 101.3 kPa.
- **Buoyancy & Archimedes' Principle (8.4):** The upward force experienced by an object immersed in a fluid is called the **buoyant force**. It is equal to the weight of the fluid displaced by the object ($F_B = \rho_{\text{fluid}} \times V_{\text{displaced}} \times g$).

8.5 - 8.6: Fluid Dynamics

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decreases, the speed of the fluid must increase to maintain a constant volumetric flow rate.

- **Bernoulli's Principle (8.5 & 8.6):** An expression of energy conservation in fluids. It states that an increase in the speed of a fluid occurs simultaneously with a decrease in static pressure (applied in airplane lift and Venturi tubes).

2018 EUEE QUESTIONS - CHAPTER 8

9. A stream of water from a wide river enters a narrower and shallow gate of a lake. Which one of the following statements is true?

A. The speed of the water decreases...

B. The speed of the water increases as it enters the gate of the lake.

C. The density of the water entering is higher...

D. The volume of water entering is higher...

Answer: B

Since the gate is narrower (Area decreases), velocity must increase ($A_1v_1 = A_2v_2$).

27. A water hose of diameter 4 cm is connected to a tap of diameter 6 cm. The water flow rate from the tap is $1.57 \times 10^{-4} \text{ m}^3/\text{s}$. What will be the speed of the water in the hose?

A. 0.125 m/s

B. 0.393 m/s

C. 0.174 m/s

D. 0.560 m/s

Answer: A

Flow Rate $Q = A \cdot v$. Area of hose = $\pi r^2 = 3.14 \cdot (0.02)^2$. $v = Q/A = 1.57\text{e-4} / 1.256\text{e-3} = 0.125 \text{ m/s}$.

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A. weight of the object.

B. turbulent force.

C. buoyant force.

D. force of the fluid.

Answer: C

Archimedes' principle defines this as the buoyant force.

56. Three different shapes of containers are filled with water at different levels... At which point does a student measure the highest pressure?

A. At point B (Deepest point)

B. At point C

C. At point D

D. At point A

Answer: A

Pressure depends purely on depth (h) via $P = \rho gh$. Container shape is irrelevant.

Chapter 9: Waves & Optics

From string vibrations to light rays, wave behavior follows strict geometric and algebraic rules.

9.1 - 9.6: Mechanical Oscillation & Sound

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also be written as $v = \lambda/T$. Waves transport *energy* without transporting matter.

- **Oscillators (9.2):** Pendulums and mass-spring systems undergo Simple Harmonic Motion (SHM). For a pendulum, Period $T = 2\pi\sqrt{L/g}$.
- **Sound Waves (9.4 & 9.6):** Sound is a longitudinal mechanical wave requiring a medium. The pitch corresponds to frequency, and loudness corresponds to amplitude.
- **Superposition (9.5):** When two waves overlap, their displacements add up algebraically (constructive or destructive interference, creating phenomena like beats and standing waves).

9.7 - 9.14: EM Waves & Geometrical Optics

- **EM Waves & Spectrum (9.7 & 9.8):** Electromagnetic waves do not require a medium and travel at $c = 3 \times 10^8$ m/s in a vacuum. The spectrum goes from Radio (lowest energy/longest λ) → Microwave → Infrared → Visible → UV → X-ray → Gamma (highest energy/shortest λ).
- **Reflection & Refraction (9.10):** Angle of incidence = Angle of reflection. Refraction obeys Snell's Law ($n_1 \sin \theta_1 = n_2 \sin \theta_2$).
- **Mirrors & Lenses (9.11):** In geometric optics, a light ray moving parallel to the principal axis of a converging lens will refract and pass directly through the principal focus.
- **Human Eye & Color (9.12 - 9.14):** A common eye defect is **Hypermetropia (farsightedness)**, where close objects cannot be clearly seen. It is corrected using convex (converging) lenses. Myopia (nearsightedness) is corrected with concave lenses. The primary additive colors of light are **Red, Green, and Blue (RGB)**.

2018 EUEE QUESTIONS - CHAPTER 9

Angle of refraction?

A. $\sin^{-1}(0.90)$

B. $\sin^{-1}(0.80)$

C. $\sin^{-1}(0.53)$

D. $\sin^{-1}(0.68)$

Answer: D

Snell's Law: $1.50 \cdot \sin(37) = 1.33 \cdot \sin(r)$. $1.50 \cdot 0.6 = 0.9$. $\sin(r) = 0.9/1.33 = 0.68$.

19. A traveling wave has a speed v , wavelength λ , period T and frequency f . Which equation correctly states the wave equation?

A. $v = \lambda T$

B. $f = 1/(\lambda v)$

C. $v = \lambda f$

D. $T = 1/(\lambda f)$

Answer: C

Velocity equals wavelength times frequency.

A. {red, blue, orange}

B. {blue, green, red}

C. {green, blue, black}

D. {red, green, yellow}

Answer: B

The additive primary colors of light in physics are Red, Green, and Blue (RGB).

53. For a person with defects of vision known as hypermetropia:

A. close objects cannot be clearly seen, so the defect can be corrected by concave lenses.

B. far objects cannot be clearly seen...

C. far objects cannot be clearly seen...

D. close objects cannot be clearly seen, so the defect can be corrected by convex lenses.

Answer: D

Hypermetropia = farsightedness. Can't see close objects well. Converging (convex) lenses correct it.

A. through the optical center will emerge parallel to the principal axis.

B. through the principal focus will emerge parallel to the principal axis.

C. parallel to the principal axis will emerge without any deviation.

D. parallel to the principal axis will emerge through the optical center.

Answer: B

Rays passing through the focus refract to emerge parallel to the axis.

Chapter 10: Electrostatics and Electric Circuits

Charges create electric fields at rest; moving charges create dynamic electrical circuits.

10.1 - 10.4: Electrostatics

- **Charges in Nature (10.1):** Charge is **quantized**; every charge in nature is an exact integer multiple of an electron's charge ($e = 1.6 \times 10^{-19} \text{ C}$). Charge is also conserved.
- **Methods of Charging (10.2):** Conduction (contact) results in the bodies sharing the same charge type. Induction (no contact + ground) forces the bodies to have opposite charge types. Friction involves stripping electrons from one body to another.
- **Coulomb's Law (10.3):** $F = k |q_1 q_2| / r^2$. Like charges repel, opposites attract.
- **The Electric Field (10.4):** Electric Field (E) is defined exactly as the electric *force per unit charge* that a positive test charge experiences ($E = F/q$).
- **Equipotential Lines:** Lines connecting points of equal voltage. There is ZERO potential difference along a single equipotential line, but there IS a non-zero potential difference between *different* equipotential lines. Electric field lines are always perpendicular to equipotential lines.

provides the electromotive "push".

- **Combination of Resistors (10.6):** Series resistors sum up directly ($R_{eq} = R_1 + R_2 + \dots$), sharing the same current. Parallel resistors sum reciprocally ($1/R_{eq} = 1/R_1 + 1/R_2 + \dots$), sharing the same voltage drop.
- **Electrical Safety (10.7):** Fuses, circuit breakers, and grounding wires protect circuits from overloading and protect people from shocks.

2018 EUEE QUESTIONS - CHAPTER 10

4. The amount of electric charge on a particle or object is an integral multiple of the charge of an electron. This property indicates:

A. charge creation.

B. charge induction.

C. charge quantization.

D. charge conservation.

Answer: C

Quantization means coming in discrete, indivisible packets.

equipotential lines?

- A. For point charge, lines are straight and parallel.
- B. For uniform field, lines are parallel to field lines.
- C. Electric potential difference is non-zero between two points on different equipotential lines.
- D. Electric potential difference is non-zero between two points on the same equipotential line.

Answer: C

Different lines have different voltages. The same line has zero voltage difference across it.

44. Electric field at a point is defined as the electric:

- A. force per unit mass...
- B. force that a positive test charge exerts...
- C. potential energy per unit charge...
- D. force per unit charge that a positive test charge experiences when placed at that point.

Answer: D

$E = F/q$. It is the force experienced by the test charge placed in the field.

B by induction, using identical negatively charged rods...

- A. Both spheres end up negatively charged.
- B. Both spheres end up positively charged.
- C. Sphere A ends up positively charged.
- D. Sphere B ends up positively charged. (A is negative, B is positive)

Answer: D

Conduction with negative rod makes A negative. Induction pushes electrons away, leaving B positive.

Chapter 11: Electromagnetism

Magnetism is inextricably linked to electricity. Moving charges generate magnetic fields, and changing magnetic fields induce electrical current (Faraday's Law).

11.1 - 11.4: Magnetic Fields, Earth, and Induction

- **Magnets & Fields (11.1):** Magnetic field lines always exit from the **North Pole** and enter the **South Pole**. They form continuous closed loops.
- **Earth's Magnetic Field (11.2):** The Earth behaves like a giant bar magnet. A compass aligns with these lines. Geographic North is roughly Magnetic South.
- **Current-Carrying Conductors (11.3):** A current passed through a wire generates a magnetic field that can deflect a compass needle without touching it. The direction is found using the Right-Hand Grip Rule.

Parallel Wires: When two parallel wires carry current in the **SAME** direction, they **ATTRACT** each other. This is because the magnetic field from wire 1 is opposite to the magnetic field from wire 2 at the location of the wires, producing an inward Lorentz force. Oppositely directed currents repel.

- **Applications of Magnetism (11.4):** Electromagnetic induction (generators producing electricity),

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2018 EUEE QUESTIONS - CHAPTER 11

28. Suppose you want to check, without any touch, if there is a current flowing through a wire or not. Which device is used?

A. Test light

B. Voltmeter

C. Compass

D. Ohmmeter

Answer: C

A compass needle (a small magnet) deflects in the presence of a magnetic field generated by a current.

40. Consider the magnetic field lines between two bar magnets are partially drawn... Lines go from L to R. The poles are:

A. North and South poles.

B. North and North poles.

C. South and North poles.

D. South and South poles.

Answer: A

Field lines exit North and enter South. If lines go $L \rightarrow R$, L is North and R is South.

force between the two conductors is attractive because the:

A. magnitude of magnetic field produced...

B. direction of magnetic field produced by I1 at C2 is opposite to that produced by I2 at C1.

C. magnitude of magnetic field produced...

D. direction of magnetic field produced by I1...

Answer: B

By Right Hand Grip Rule, B1 points INTO page at C2, and B2 points OUT OF page at C1. These opposite vectors produce attraction.

58. A square wire loop (side 12 cm) has area vector angle 60° with $B=10$ T. Pulled out in 6 s. Induced EMF?

A. 24 mV

B. 17.2 mV

C. 12 mV

D. 20.9 mV

Answer: C

Flux = $BA\cos(60) = 10 * (0.12)^2 * 0.5 = 0.072$ Wb. EMF = $\Delta\Phi/\Delta t = 0.072 / 6 = 0.012$ V = 12 mV.

Chapter 12: Electronics

Modern circuits rely heavily on semiconductor materials (silicon/germanium) whose conductive properties can be manipulated via "doping"

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- **Semiconductors (12.1):** Intrinsic (pure) semiconductors are poor conductors. Doping creates Extrinsic semiconductors: P-type (positive hole charge carriers) and N-type (negative electron charge carriers).
- **Diodes (12.2) & Rectification (12.3):** A PN junction forms a **diode**. It acts as a one-way valve: it conducts significant current ONLY in forward bias, where the p-side acts as the Anode (+) and the n-side acts as the Cathode (-). Diodes are primarily used in **rectification** (converting AC current to DC current).
- **Transistors (12.4):** Three-layer devices (NPN or PNP) used as amplifiers or switches. In a schematic symbol for an **NPN transistor**, the arrow on the emitter points **OUTWARD** (memory trick: NPN = Not Pointing iN). For a PNP transistor, the arrow points inward.
- **Integrated Circuits & Logic Gates (12.5 - 12.7):** Logic gates are the building blocks of digital electronics based on Boolean logic. Major gates include AND, OR, NOT, NAND, NOR.

2018 EUEE QUESTIONS - CHAPTER 12

26. Which of the following is a correct schematic symbol for an NPN transistor?

A. Emitter arrow pointing OUT

B. Emitter arrow pointing IN

C. Emitter pointing IN

D. Emitter pointing IN

Answer: A

NPN = Not Pointing iN. The arrow must point OUT.

A. that conducts significant current in both forward and reverse biasing.

B. that will have an ohmic type current-voltage characteristics.

C. where a p-side is anode and the n-side is cathode at the junction.

D. that conducts current in the reverse biasing...

Answer: C

A diode is forward biased (conducts) when positive (Anode) goes to P-type and negative (Cathode) goes to N-type.

Chapter 13: Nuclear Physics and Radiation

The atom consists of an incredibly dense nucleus surrounded by electron clouds. Nuclear physics deals with interactions deep within this nucleus.

13.1 - 13.5: The Nucleus, Radioactivity & Reactions

- **The Nucleus (13.1):** Contains protons and neutrons (collectively called nucleons). The **Strong Nuclear Force** is a strictly short-range, incredibly powerful attractive force that acts between nucleons inside the nucleus, overcoming their massive electrostatic repulsion.
- **Radioactivity (13.2):** Unstable nuclei spontaneously decay by emitting Alpha (α , helium nucleus), Beta (β , electron/positron), or Gamma (γ , high-energy photon) radiation. Decay is characterized by a **half-life ($T_{1/2}$)**, which is the time taken for half the radioactive nuclei in a sample to decay. The decay constant (λ) is found via: $\lambda = \ln(2) / T_{1/2}$.
- **Use of Nuclear Radiation (13.3):** Isotopes are used in medicine (tracers, cancer therapy), agriculture, and industry (thickness gauges, flaw detection).
- **Nuclear Reactions & Energy (13.4):** *Fission:* Splitting a heavy nucleus into lighter ones (used in nuclear power plants). *Fusion:* Combining light nuclei into a heavier one (powers the sun and stars).

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minimizing exposure time, and utilizing dense shielding (like lead or thick concrete).

2018 EUEE QUESTIONS - CHAPTER 13

20. The iodine isotope ^{131}I which has an eight-day half-life, is used in nuclear medicine. What is the decay constant of the isotope? ($\ln 2 = 0.693$)

A. 5.64 / day

B. 8.66×10^{-2} / day

C. 1.17×10^{-2} / day

D. 12.5 / day

Answer: B

$$\lambda = 0.693 / 8 = 0.086625 \text{ / day} = 8.66 \times 10^{-2} \text{ / day.}$$

31. A strong nuclear force is:

A. an attractive force between the nucleus and the electrons...

B. a short-range attractive force that acts between nucleons in the nucleus.

C. a short-range repulsive force...

D. a short-range force which acts inside the individual nucleons...

Answer: B

It binds protons and neutrons (nucleons) together in the nucleus at very short ranges.



Best of Luck with your Exams!



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